

3. Nuclear fusion can take place in a fusion reactor.

- (a) Deuterium is one isotope of hydrogen, H, which consists of 1 proton and 1 neutron.
Tritium is another isotope of hydrogen which consists of 1 proton and 2 neutrons.
In one nuclear fusion reaction, deuterium and tritium undergo nuclear fusion to form helium, He, and one neutron.

Produce a **balanced nuclear equation** for the fusion of deuterium and tritium. [3]

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- (b) (i) Explain why nuclear fusion is difficult to achieve on Earth. [2]

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- (ii) One advantage of nuclear fusion compared to nuclear fission is that it doesn't produce radioactive waste.
State **two** reasons why radioactive waste is difficult to store safely. [2]

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